IF EVERY QUOTIENT BY A PRIME IS J-0 THEN THE RING IS J-1

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The criterion for a Noetherian ring to be J-1. Lemma \ref{lemma-J-1}, cited at proof lines 11624-11625. Is the statement that is verified at each regular prime.
- Elements generating the maximal ideal of a regular local ring form a regular sequence. Algebra, Lemma \ref{algebra-lemma-regular-ring-CM}, cited at proof lines 11629-11631. Gives the regular sequence in the localisation.
- A regular sequence can be spread out to a neighbourhood. Algebra, Lemma \ref{algebra-lemma-regular-sequence-in-neighbourhood}, cited at proof lines 11632-11634. Allows replacing the ring by a localisation in which the chosen elements form a regular sequence and generate the prime.
- A local ring is regular when it is the quotient of a regular local ring by a regular sequence. Algebra, Lemma \ref{algebra-lemma-regular-mod-x}, cited at proof lines 11642-11644. Concludes regularity at the prime in question.

Source reference. Original lemma 07P9: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9789–9789. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 07P8: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9788–9788. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

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Source reference. Original lemma 00NQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 868–868. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 061L: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7765–7765. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00NU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 872–872. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let R be a Noetherian ring. Let $X = \mathrm{Spec}(R)$. Assume that for all primes $\mathfrak{p} \subset R$ the ring $R/\mathfrak{p}$ is J-0. Then R is J-1.*

Proof. We will show that the criterion of Lemma 07P8 applies. Let $\mathfrak{p} \in \mathrm{Reg}(X)$ be a prime of height r. Pick $f_1, \ldots, f_r \in \mathfrak{p}$ which map to generators of $\mathfrak{p}R_\mathfrak{p}$. Since $\mathfrak{p} \in \mathrm{Reg}(X)$ we see that $f_1, \ldots, f_r$ maps to a regular sequence in $R_\mathfrak{p}$, see Algebra, Lemma 00NQ. Thus by Algebra, Lemma 061L we see that after replacing R by R_g for some $g \in R$, $g \notin \mathfrak{p}$ the sequence $f_1, \ldots, f_r$ is a regular sequence in R. After another replacement we may also assume $f_1, \ldots, f_r$ generate $\mathfrak{p}$. Next, let $\mathfrak{p} \subset \mathfrak{q}$ be a prime ideal such that $(R/\mathfrak{p})_\mathfrak{q}$ is a regular local ring. By the assumption of the lemma there exists a non-empty open subset of $V(\mathfrak{p})$ consisting of such primes, hence it suffices to prove $R_\mathfrak{q}$ is regular. Note that $f_1, \ldots, f_r$ is a regular sequence in $R_\mathfrak{q}$ such that $R_\mathfrak{q}/(f_1, \ldots, f_r)R_\mathfrak{q}$ is regular. Hence $R_\mathfrak{q}$ is regular by Algebra, Lemma 00NU. $\square$